AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter Monitoring & Health Effects

Issue 2026-08-01 | Entry window 1 August 2026, plus first sweep of the new by-journal leg | 25 records in scope

25 RECORDS IN SCOPE (8 SUBTOPICS)	**9** SENSING / INSTRUMENTATION	**17** FULL ENTRIES (8 METADATA-ONLY)	**5** RECORDS FROM CHINA (20% OF CORPUS)	**2** EFFECT ESTIMATES EXTRACTED

Signal of the day

Incense arrives from both ends on the same day: a birth cohort finds a dose–response for small-for-gestational-age, and a chamber study shows why a mass-based monitor would have missed the exposure entirely.

Pai et al. (*Pediatr Neonatol*) analysed **18,666 mother–infant pairs** in the Taiwan Birth Cohort Study with household incense exposure categorised as none, occasional or daily. Among term infants, SGA odds were **aOR 1.16 (95% CI 1.00–1.33)** for occasional and **1.25 (1.10–1.41)** for daily burning — an ordered dose–response, robust to adjustment for other indoor sources, maternal medical history, gestational weight gain and secondhand smoke, with no association for preterm birth. Exposure is self-reported and unquantified, which is the study's real limit: it establishes that the practice matters without saying what dose does.

Zhou et al. (*Atmos. Chem. Phys.*) supply the missing half. Chamber characterisation of Arabian incense (Bakhoor) under charcoal heating gives emission rates of **670–1690 μg min^{-1} g^{-1}** by mass and **(6–7)$\times$10^{11} particles min^{-1} g^{-1}** by number, with **ultrafine particles contributing 70–75% of total particle number** and UFP emission rates exceeding sidestream cigarette smoke. Water-soluble DTT activity was $\sim$32 pmol min^{-1} μg^{-1} — modestly *below* cigarette particles — yet the intracellular oxidative-stress response was *stronger*, and ozone ageing raised oxidative potential for both sources.

Read together the operational point is unambiguous. The health signal sits in a source whose emission is dominated by particle *number* in the ultrafine range, whose toxicity ranking inverts between acellular and cellular assays, and whose burden is therefore invisible to the $PM_{2.5}$-mass instrument that every relevant epidemiological study — including Pai's — would have deployed. Number-resolved, low-cost indoor sensing is not a refinement here; it is the only way to put an exposure metric on the association.

What changed today

- **The sensing blackout is over, and it was a harvest defect exactly as diagnosed.** The 31 July issue recorded a second consecutive window with zero sensing records and prescribed the fix: a Crossref by-journal feed keyed on `created`. That leg was built and run today across eight journals PubMed does not index or indexes late. It returned **55 deposits** for 26 July–1 August, of which 12 were PM-relevant. Today's issue carries **9 sensing/instrumentation records** against 0, 0 and 1 in the three preceding issues. The defect was never a real absence in the literature.
- **Calibration papers now agree on what breaks, and it is not the sensor.** Kumpika et al. reach $R^2 = 0.974$ against a BAM with a third-order humidity correction *and* report that duty-cycling a PMS7003 to 20 min/day slows drift relative to continuous operation; Qian et al. halve NRMSE with AutoML but only by splitting models across clean-air and pollution-event concentration ranges; Le et al. trace a ceilometer's signal loss to window fogging from insufficient firmware-controlled heating, and show the internal calibration factor stops compensating once laser power falls below 40%. Three different instrument classes, one conclusion: the dominant error terms are environmental and ageing-related, and they are not on the manufacturer's specification sheet.
- **Exposure-model methodology got two hard results on the same failure mode.** Saeedi et al. show land-use regression for mobile ultrafine particles collapsing from **$\sim$217% average percentage error under random cross-validation to $\sim$79% under spatiotemporal CV with forward feature selection** — i.e. the standard validation protocol was reporting a model that could not predict. Blanco et al. then resample on-road monitoring designs against a 5,283-person cohort and find health-effect estimates attenuate under every alternative design, with short 4-visit campaigns the most unstable. Both papers say the same thing: reported LUR performance is a property of the validation scheme, not of the exposure surface.
- **Two independent suicide records surfaced in one window.** Daoud et al. report a fixed-effects panel association across UK local authorities ($\beta = 0.221$, $p = 0.007$) with a U-shaped form turning at $\approx$7.8 μg/m^3; Cai et al. (*Atmos. Environ.*, metadata only) report $PM_{2.5}$ as the dominant pollutant in a province-wide case-crossover among people living with HIV/AIDS in China. Neither is strong evidence alone — Daoud's is ecological with no time-varying socioeconomic controls — but the convergence marks a subtopic that is accumulating.

- **Source attribution keeps demoting the sector everyone regulates first.** Zhou et al.'s 2022 India WRF-Chem run (annual bias $0.2 \pm 16.9\,\mu\mathrm{g/m}^3$ across 288 sites, population-weighted mean $47.4\,\mu\mathrm{g/m}^3$) finds **transboundary transport contributing 27%** — more than any single domestic sector — with industry at 18% now ahead of residential at 15% nationally. The residential decline is a real policy result; the transboundary share is a jurisdictional problem no national instrument reaches.
- **Oxidative potential became a modelled quantity.** Mishra et al.'s KM-OP kinetic model reproduces measured OP in Grenoble, Paris and London at $R^2 > 0.75$ with $< 15\%$ bias in 4 of 6 datasets, identifies organic aerosol as the strongest contributor to antioxidant-based assays, and separates Cu (dominates $\mathrm{H_2O_2}$) from the •OH production that actually tracks the assays. This is the first record in this watch that lets OP be predicted from composition rather than measured per filter.

Corpus analytics

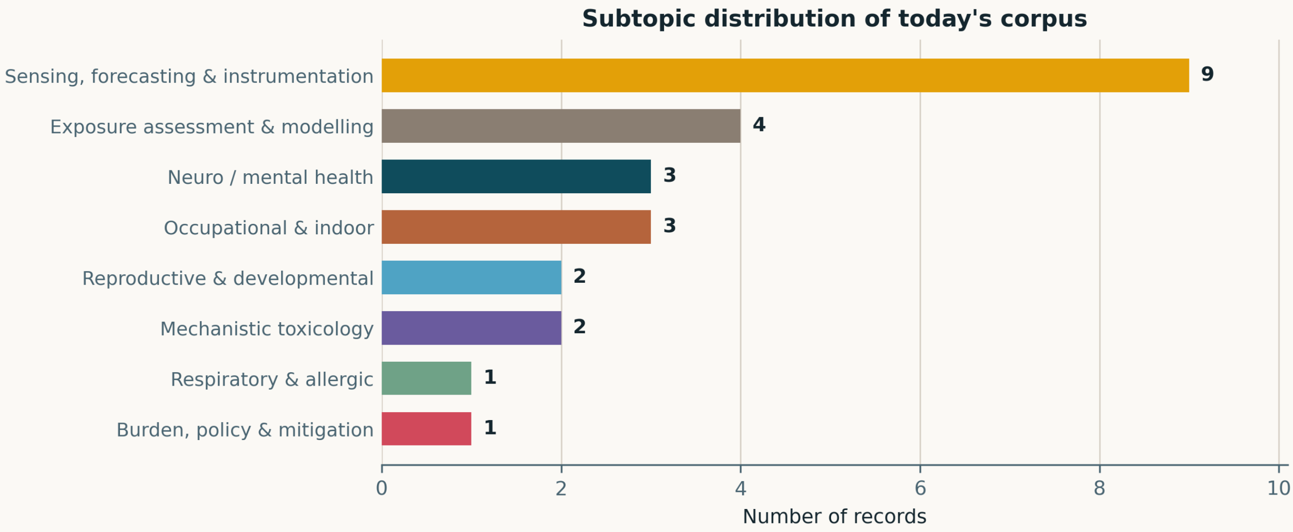

Fig. Subtopic assignment of the 25 in-scope records, single-label by dominant contribution. *Sensing, forecasting & instrumentation* is the largest cluster at 9 records — the first time it has led an issue — and is entirely a product of the by-journal leg activated today. Exposure assessment follows at 4. The health clusters are correspondingly thin because the PubMed entry window covers a single day (1 August) while the sensing leg back-swept a week.

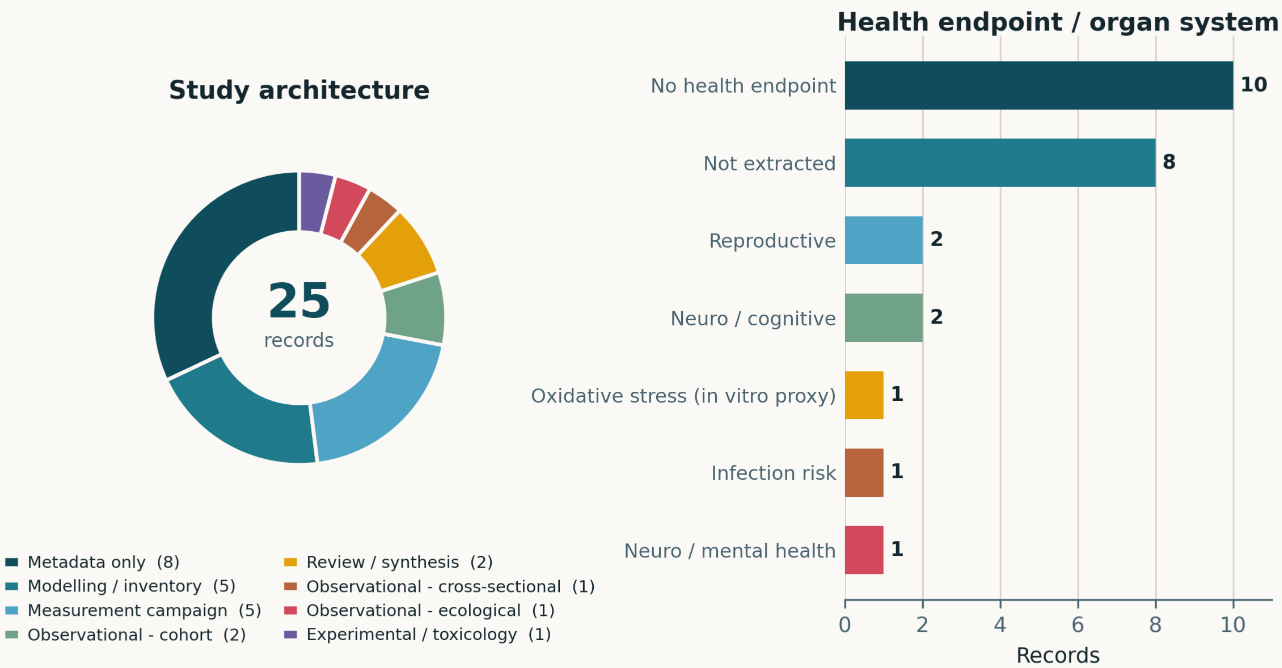

Fig. Left — study architecture. Eight records are metadata-only (Elsevier titles whose abstracts are deposited in neither Crossref nor Europe PMC at this point in the publication cycle) and are given their own slice rather than an inferred design; the remaining 17 span modelling/inventory, measurement campaign, review, cohort, cross-sectional, ecological and experimental architectures. Right — health endpoint by organ system. **Ten records carry no human health endpoint at all and eight more were not opened**, which is what an instrumentation-weighted issue looks like. The endpoints that do appear are neuro/cognitive (2), neuro/mental health (1), reproductive (2), infection risk (1) and an in-vitro oxidative-stress proxy (1).

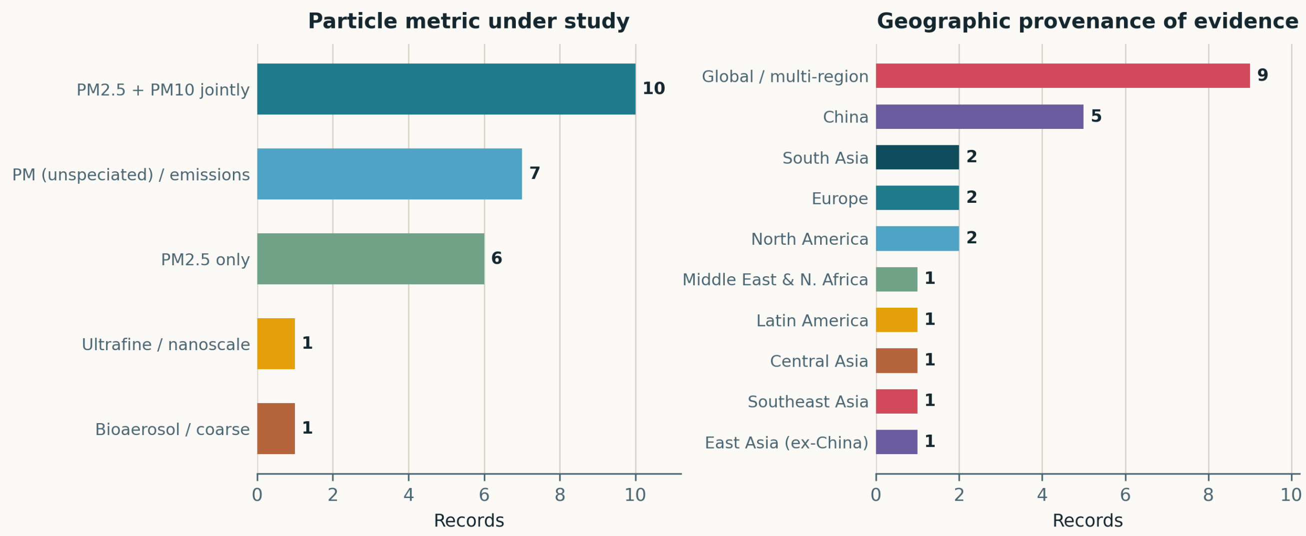

Fig. Left — particle metric. **Ultrafine particle number appears in three records today** (Bakhoor emissions, Toronto mobile LUR, Seattle on-road design) against a recent baseline of zero or one — the direct consequence of reading the aerosol-science journals rather than only the biomedical ones. Right — geographic provenance. China leads at 5 (20%), well below its usual share, and chamber and simulation studies carry no national attribution.

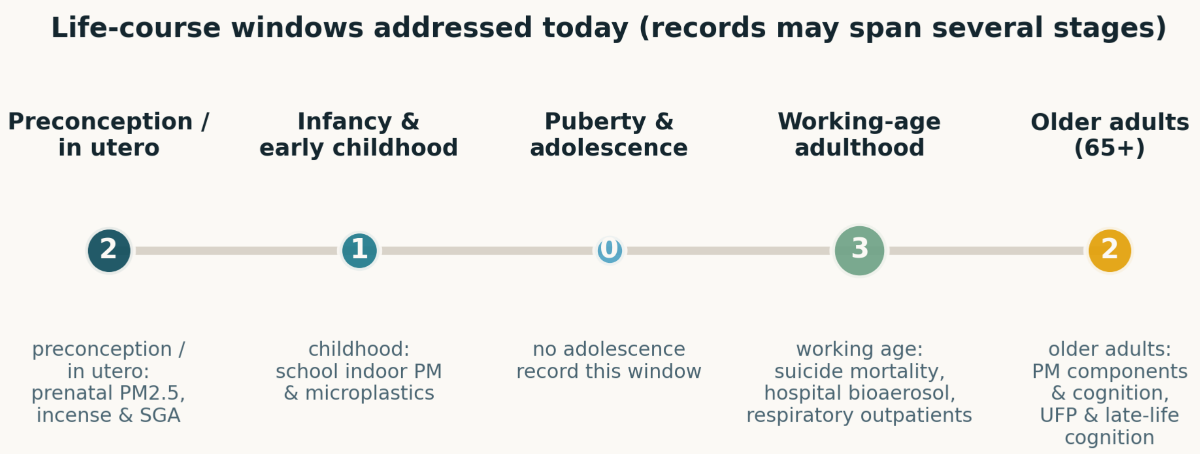

Fig. Life-course windows addressed; counts are non-exclusive and only records with an identifiable human population are counted. Working age leads at 3 (suicide mortality, hospital bioaerosol exposure, school indoor exposure), with the in-utero window at 2. Adolescence is empty again.

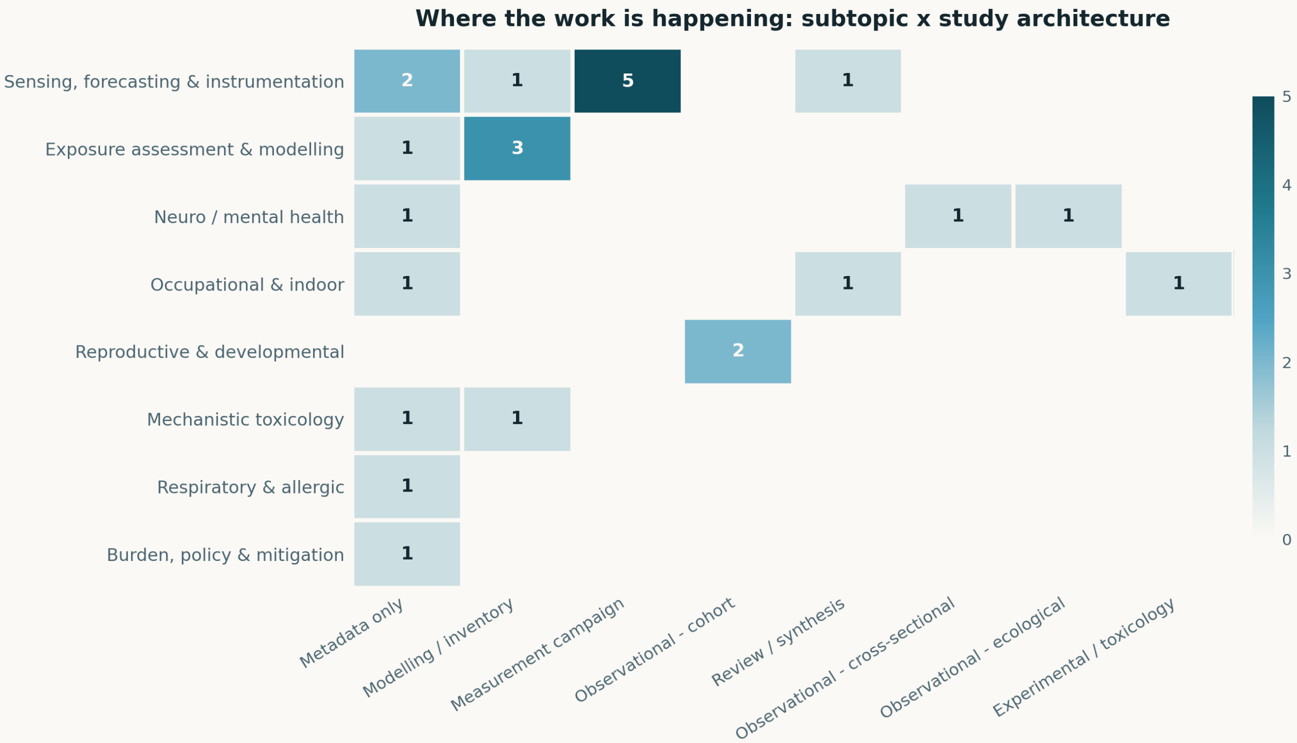

Fig. Subtopic × study-architecture cross-tabulation. The structural feature to note is that the sensing row is now populated across four distinct architectures — field co-location, chamber, network deployment and simulation — rather than the single forecasting-model cell it collapsed to in previous issues. The health rows remain observational-only; no record today links an instrument characterisation to a health endpoint within the same study.

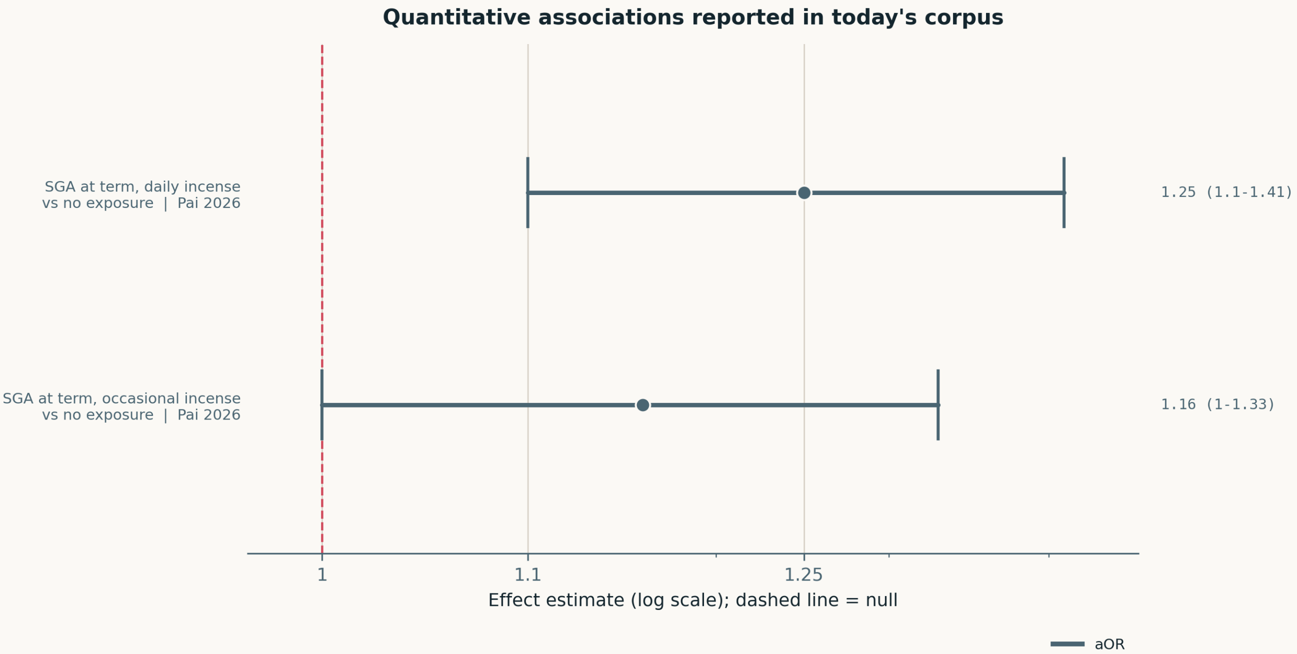

Fig. The two quantitative effect estimates in the corpus, both from Pai et al.'s incense–SGA analysis, on a common logarithmic axis. **Two estimates is an unusually thin forest and reflects the issue's composition rather than the literature**: instrumentation and source-attribution papers report performance metrics, not risk ratios, and the eight metadata-only records were not opened. Zhang et al. report odds increases of 19–43% per IQR across six $PM_{2.5}$ components but give no per-component interval in the abstract; Bello et al. report -6.6 g birth weight per 1 $\mu g/m^3$ in the most vulnerable stratum with no published CI; Daoud et al. report a panel coefficient. None of the three is plottable without inventing an interval, so none was plotted.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Eight records surfaced without a retrievable abstract are listed separately at the end on metadata alone — they are not summarised, because there is nothing to summarise from.

Sensing, forecasting & instrumentation 5 full entries

Kumpika et al. 2026 Environ Pollut Bioavailab

Correction of humidity and sensor ageing effects in low-cost $PM_{2.5}$ light-scattering sensors

Field campaign in Chiang Mai with PMS7003 units across a wide relative-humidity range, validated against a government BAM and cross-checked with dry-air measurement through a diffusion dryer. Polynomial corrections up to third order were fitted; **agreement with BAM reached** $R^2 = 0.9737$ after correction, with the largest gains at elevated RH. The second result is the more interesting one: **restricting operation to 20 min/day measurably reduced long-term drift** relative to continuous running, implicating cumulative optical-chamber exposure rather than elapsed calendar time as the ageing mechanism. Single-site, single sensor model, single season, so the polynomial coefficients are not transferable — but duty-cycling is a design lever, not a calibration constant, and it applies to any continuously powered node.

doi: 10.1080/26395940.2026.2616098

Qian et al. 2026 Atmos Meas Tech

Enhancing accuracy of indoor air-quality sensors via automated machine-learning calibration

Multi-stage AutoML calibration chaining field sensors to intermediate drift-correction reference sensors to a reference-grade instrument, with **separate models fitted below and above a concentration break** for clean air versus pollution events. Evaluated in a controlled indoor chamber with two sensor models against diverse real indoor sources under uncontrolled ambient conditions: $R^2 > 0.90$**, NRMSE and sMAPE roughly halved** against uncalibrated output, bias effectively removed. The split-range design is the transferable idea — a single global calibration is being asked to serve two distinct optical regimes and it fails at both ends. Chamber conditions make the result an upper bound on field performance, and no ageing term is included, which the Kumpika result suggests is the dominant long-run error.

doi: 10.5194/amt-19-603-2026

Le et al. 2026 Atmos Meas Tech

Operational performance of the Vaisala CL61 ceilometer for atmospheric profiling

Three years of operational data from multiple CL61 units — a depolarisation-capable elastic backscatter lidar — assessed for inter-unit consistency, drift and bias. Most early production units showed **progressive laser-power decline with rising background noise**, and manual calibration against atmospheric targets at Lindenberg showed that **once laser power falls below 40% the internal calibration factor stops compensating**. Instrumental background and bias vary with temperature; a correction and uncertainty-estimation method is given. Under low aerosol load the bias correction shifts volume linear depolarisation ratio by up to 0.005, while the volume-to-particle depolarisation difference reaches 0.1 — so molecular correction, not instrumental bias, dominates quantitative aerosol interpretation at 910.55 nm. One unit's signal loss was traced to window fogging from insufficient firmware-driven heating. Instructive as a template: silent, firmware-mediated degradation is exactly the failure mode a distributed network cannot detect without co-location.

doi: 10.5194/amt-19-4923-2026

Silva et al. 2026 J Air Waste Manag Assoc

Low-cost sensors for indoor air-quality monitoring: systematic review of accuracy, applications and limitations

PRISMA review, **24 studies retained from 2,736 screened**, covering optical particle counters and NDIR CO_2 against reference-grade instruments in real settings. Strong correlations are achievable under controlled or semi-controlled conditions, but performance degrades with humidity, temperature and source type, and the authors flag methodological heterogeneity, near-absent inter-device validation, regional bias and **unaddressed sensor ageing and cross-sensitivity** as systematic gaps. The operational conclusion — that most devices do not meet regulatory compliance requirements and standardised validation protocols do not exist — is well supported and worth citing directly. As a review it inherits the heterogeneity it documents, and a 24-study base is thin for the breadth of the claims.

doi: 10.1080/10962247.2026.2624795

Nurseitov et al. 2026 Information

Design, deployment and field evaluation of a low-cost IoT monitoring network for urban particulate matter

Six custom nodes (Winsen ZPHS01B on Raspberry Pi Zero 2 W, ~US$100/node) deployed across Almaty over a February–April campaign, >**70,000 measurements**, with three engineering contributions: an active airflow-stabilisation enclosure decoupling sampling from wind, context-aware adaptive edge filtering cutting transmitted volume ~40%, and a segmented telemetry pipeline. Cross-validation was against a consumer monitor with documented traceability rather than a reference analyser ($R^2 = 0.89$–0.95), and the authors state explicitly that the network provides *internally consistent* rather than reference-equivalent concentrations — unusually honest framing. **Daily mean $PM_{2.5}$ exceeded the WHO 24 h guideline on 84% of days** (February 54.4 vs March 21.9 $\mu g/m^3$); a $PM_{2.5}/PM_{10}$ ratio near 0.96 plus higher weekend levels points to residential coal heating. That ratio is high enough to warrant suspicion of the coarse channel rather than confidence in the source inference.

doi: 10.3390/info17070642

Sensing — lower-tier records 2 entries

Reis 2026 Sensors

Low-power embedded sensor node for environmental monitoring with on-board machine-learning inference

Architecture and system-level optimisation of a multi-parameter node coupling a low-power microcontroller to a neural inference accelerator, with adaptive duty-cycling, hierarchical power domains and quantised fixed-point models doing three-class normal/anomalous/critical classification at the edge. The reported figures — **94% inference accuracy, 0.87 ms latency, ≈2.9 mWh average power, ≈88% reduction in LoRaWAN transmissions** — come from MATLAB/Simulink simulation, not deployment, and there is no co-location against a reference instrument anywhere in the study. Treat the energy budget as a design target and the accuracy figure as unvalidated. The duty-cycling and hierarchical-power argument is the reusable part, and it is also the part that does not depend on the classifier working.

doi: 10.3390/s26020703

Mohapatra et al. 2026 IEEE Access

IoT–machine-learning proof-of-concept framework for real-time urban particulate matter prediction

PMS7003 on an Arduino module at campus scale, with outlier treatment, normalisation and temporal indexing feeding random-forest and polynomial regression, plus k-means clustering to identify persistent high-exposure zones. Predictive performance **exceeds 85% for $PM_{1.0}$ and $PM_{2.5}$ but degrades sharply at 5 and 10 μm**, which the authors correctly attribute to discretised, low-variability sensor output rather than to the model — a useful negative result about the coarse channels of optical counters generally. No reference co-location, single site, and the models predict the sensor's own future readings rather than true concentration, so "performance" here is self-consistency. Included for the size-channel finding only.

doi: 10.1109/access.2026.3651996

Exposure assessment & modelling — 3 full entries

Saeedi et al. 2026 Environ Sci Technol

Feature selection and cross-validation strategy in land-use regression for mobile-monitored ultrafine particles

Toronto mobile UFP campaign used to compare LUR models trained under random, spatial, temporal and spatiotemporal cross-validation, with and without forward feature selection, each evaluated on a hold-out test set and against *independent* stationary backyard measurements. **Average percentage error fell from ∼217% under random CV to ∼79% under spatiotemporal CV with FFS.** Random-CV models overfitted, were sensitive to outliers, produced spatial artefacts and exaggerated variable effects, and failed on independent samples — while reporting good internal validation statistics. Spatial folds were designed at scales reflecting UFP autocorrelation, which is the methodological core. The blunt implication: a large share of published LUR performance numbers are measuring data reproduction, and any exposure surface fed into an epidemiological model should carry a spatiotemporally validated error, not a random-CV one.

doi: 10.1021/acs.est.5c12601

Blanco et al. 2026 J Expo Sci Environ Epidemiol

Influence of on-road mobile monitoring design on ultrafine particle exposure models and cognitive health inferences

A reference roadside UFP model built from repeated measurements at 309 locations was used with the Adult Changes in Thought cohort ($n = 5{,}283$) to test late-life cognitive function, then **600 km of on-road measurements were subsampled** to emulate alternative campaign designs varying visit frequency, spatial balancing and sampling time, with universal kriging plus partial least squares. The reference model gave a null: **CASI-IRT** $+0.007$ **(95% CI** -0.013 **to** 0.027**) per 1,900 pt/cm^3**. Alternative designs reproduced the null but attenuated it; route-based sampling and very short 4-visit campaigns produced the most variable estimates, while temporal and plume adjustments changed little. Testing design sensitivity against a null limits what can be concluded — attenuation toward zero is unobservable when the reference is zero — and the authors say so. The design guidance, that visit count and out-of-hours coverage matter more than post-hoc adjustment, is what carries.

doi: 10.1038/s41370-026-00845-y

Zhou M et al. 2026 Atmos Chem Phys

Surface $PM_{2.5}$ in 2022 India: emission updates, WRF-Chem evaluation and source attribution

First WRF-Chem evaluation and source attribution for India for 2022, with an updated residential inventory reflecting the household clean-fuel transition, a plant-level coal-power inventory, a revised secondary-organic-aerosol scheme and improved near-surface mixing. **Annual $PM_{2.5}$ bias** $0.2 \pm 16.9\,\mu\mathrm{g/m^3}$ $(0 \pm 31\%)$ **across 288 South Asian monitoring sites**; simulated population-weighted annual mean **47.4 μg/m^3**. Attribution: transboundary transport 27% ($12.8\,\mu\mathrm{g/m^3}$) — larger than any single domestic sector — industry 18% ($8.6\,\mu\mathrm{g/m^3}$), residential 15% (7.3, still dominant in the Indo-Gangetic Plain), power 13% (6.1, concentrated in central India). A ±31% annual bias is respectable for a CTM but is the same order as the sectoral shares being separated, so the ranking of industry above residential should be read as provisional. The residential demotion is nonetheless a policy-relevant result about a transition that actually happened.

doi: 10.5194/acp-26-10533-2026

Neuro, cognition & mental health — 2 full entries

Zhang et al. 2026 Neurotoxicology

$PM_{2.5}$ and its components and cognitive dysfunction among rural older adults

14,379 Henan Rural Cohort participants aged $\geq$60, cognitive dysfunction defined by MMSE and the Hasegawa Dementia Scale, with $PM_{2.5}$, black carbon, organic matter, ammonium, sulfate and nitrate from the TAP dataset; logistic and weighted quantile sum regression, then network toxicology and random forest for candidate genes. **Per-IQR increases raised CD odds by 19–43% across the six exposures**, and **BC was the leading mixture contributor under all four CD definitions** (weights 54–86%), with OM second (7–35%). Cross-sectional, so reverse causation through cognitively impaired participants' residential and behavioural patterns is unaddressed, and component fields from a single reanalysis product are strongly collinear — WQS weights under collinearity are unstable. The BC ranking is consistent with the emerging component literature; the network-toxicology arm is hypothesis-generating and labelled as such by the authors.

doi: 10.1016/j.neuro.2026.103539

Daoud et al. 2026 Environ Sci Pollut Res Int

Air pollution and suicide mortality in the UK

Balanced panel of >**2,200 local-authority-year observations, 2015–2023**, linking ONS suicide data to DEFRA modelled $PM_{2.5}$ with local-authority and year fixed effects, plus nonlinear specifications. The raw cross-sectional association is *negative*; after fixed effects it reverses to $\beta = 0.221$ ($p = 0.007$), with a U-shaped form turning at $\approx$7.8 $\mu g/m^3$. That sign reversal is the whole result and it is not reassuring: the estimate is driven entirely by within-area variation over a period of falling concentrations and rising suicide rates, with no time-varying socioeconomic controls (deprivation, unemployment, service provision) in the model. Ecological, modelled exposure, no individual-level confounders. The authors describe it as exploratory, which is the correct reading. Logged here because a second, independent PM–suicide record surfaced in the same window (Cai et al., below).

doi: 10.1007/s11356-026-38089-w

Reproductive & developmental — 2 full entries

Pai et al. 2026 Pediatr Neonatol

Household incense burning and small-for-gestational-age risk

18,666 mother–infant pairs in the Taiwan Birth Cohort Study, incense exposure during pregnancy categorised none / occasional / daily, SGA stratified by term and preterm. SGA prevalence rose monotonically across exposure groups (8.1%, 9.4%, 10.0%). Among term infants, **aOR 1.16 (1.00–1.33) occasional and 1.25 (1.10–1.41) daily**; preterm infants showed the same direction without significance, and there was no association with preterm birth itself. Sensitivity analyses covered other indoor environmental factors, maternal medical history, gestational complications, gestational weight gain and secondhand-smoke interaction. The limitation that matters is that exposure is a self-reported frequency category with no concentration, duration or ventilation information — the dose–response is ordinal in *practice*, not in dose, and the contrast between an unventilated daily burn and a well-ventilated one sits entirely inside the top category. Pairs directly with Zhou L et al. in this issue.

doi: 10.1016/j.pedneo.2026.06.003

Bello et al. 2026 Econ Hum Biol

Prenatal $PM_{2.5}$ exposure and infant health in Quebec

Administrative birth records for Quebec 2008–2015 linked to monitoring data and tax records, in neighbourhood-by-quarter fixed-effects models — a low-pollution setting, which is the point. **Population-average effects on birth outcomes were essentially null**; the result lives entirely in the heterogeneity. Among families in the most vulnerable stratum of a cumulative socioeconomic vulnerability index, a **1 $\mu g/m^3$ increase in mean $PM_{2.5}$ was associated with $\approx$6.6 g lower birth weight**, with further effects on prematurity and birth weight among newborns admitted to neonatal care. **Days exceeding the Canadian 24 h standard had disproportionately large effects in vulnerable families**, so peaks matter beyond the mean — directly relevant to any monitoring design optimised for annual averages. No confidence intervals are reported in the abstract, and the vulnerability index is constructed rather than pre-registered, which invites specification search across strata.

doi: 10.1016/j.ehb.2026.101631

Occupational & indoor exposure — 2 full entries

Zhou L et al. 2026 Atmos Chem Phys

Indoor burning of Arabian incense generates abundant ultrafine particles with strong oxidative potential

Chamber characterisation of Bakhoor under charcoal-assisted heating representative of household practice, with acellular DTT activity and a macrophage intracellular oxidative-stress assay, benchmarked against protocol-matched sidestream cigarette smoke. Normalised per gram burned: **670–1690 $\mu g\,min^{-1}\,g^{-1}$ mass and (6–7)$\times10^{11}$ particles $min^{-1}\,g^{-1}$ number, with ultrafines 70–75% of number** and UFP emission rates above sidestream cigarette smoke. Emission magnitude followed a nonlinear power law in the hexane-soluble fraction, identifying a compositional control. **Water-soluble OP_{DTT} was $\approx$32 pmol $min^{-1}\,\mu g^{-1}$ — modestly below cigarette particles — yet the cellular response was stronger**, and ozone ageing raised OP for both sources at ageing equivalent to days of indoor residence. The acellular/cellular divergence is the finding with teeth: DTT alone would have ranked this source as the safer of the two. Chamber conditions and a charcoal co-source mean the absolute rates are not a household exposure estimate.

doi: 10.5194/acp-26-10587-2026

van Rheenen et al. 2026 J Hosp Infect

Microbial contamination of air-conditioning and humidifying systems in hospitals and the built environment

Mapping review of pathogen presence, biofilm formation and dissemination in the air and water compartments of air-handling systems analogous to adiabatic humidifiers, searched across three databases: **22 of 468 studies retained, covering 184 air-handling units, none of which used adiabatic humidification**. *Pseudomonas aeruginosa* and *Burkholderia cepacia* appeared occasionally in system air; *Mycobacterium* and *Acinetobacter* on cooler surfaces and in drain pans; *Mucor* and *Aspergillus fumigatus* in water and surrounding air, *Fusarium* more airborne. **No included study examined biofilm** — the proposed mechanism is therefore unevidenced within the review's own base. The conclusion is a statement about absence of evidence for the technology actually being adopted on energy-efficiency grounds, which is a legitimate thing to publish. Bioaerosol rather than mass concentration, so it sits at the edge of this watch's scope; included because humidification is a growing indoor-air intervention with no monitoring standard.

doi: 10.1016/j.jhin.2026.07.020

Mechanistic toxicology 1 full entry

Mishra et al. 2026 Atmos Chem Phys

Aerosol oxidative potential and reactive species predicted with a chemical kinetics model (KM-OP)

A chemical kinetic model of oxidative potential built on literature rate coefficients and a large compilation of laboratory data for transition-metal ions, quinones and organic aerosol, predicting ROS production and consumption of ascorbic acid and DTT from PM composition. Applied to field measurements from **Grenoble, Paris and London: $R^2 > 0.75$ with $< 15\%$ model bias in 4 of 6 datasets**. **Organic aerosol is the strongest contributor to antioxidant-based OP assays**, with minor Cu and Fe contributions; Cu dominates H_2O_2 production but barely affects •OH, while the antioxidant assays correlate well with modelled •OH — separating the species that drives the assay from the species that drives the metal signal. Two of six datasets are not reproduced, and the model is calibrated on the same body of literature it is tested against in kind. Even so, this converts OP from a per-filter measurement into a quantity predictable from speciated composition, which is exactly what a composition-resolving sensor network would need to be worth deploying.

doi: 10.5194/acp-26-10605-2026

Surfaced without abstract — metadata only

Eight records reached this issue through the Crossref by-journal leg but have no abstract deposited in Crossref and are not yet in Europe PMC or PubMed. They are recorded here on title, journal, deposit date and DOI alone. **No summary is offered because none can be written without inventing content**; several are clearly relevant and should be revisited once abstracts appear.

- **Cai et al.** — $PM_{2.5}$ as the dominant air pollutant associated with suicide among people living with HIV/AIDS: a province-wide case-crossover study in China. *Atmos Environ*, deposited 29 July. `10.1016/j.atmosenv.2026.122271` [pairs with Daoud et al.]
- **Lu et al.** — Antagonistic effects of $PM_{2.5}$ and ozone on respiratory morbidity: multi-pollutant interaction analysis using outpatient data from the Pearl River Delta. *Atmos Environ*, deposited 30 July. `10.1016/j.atmosenv.2026.122270` [antagonism, not synergy]
- **Zheng et al.** — Augmented transformer framework for quantification of $PM_{2.5}$ chemical composition via one-step laser desorption/ionisation single-particle aerosol mass spectrometry. *Atmos Environ*, deposited 28 July. `10.1016/j.atmosenv.2026.122260` [speciation from SPAMS — directly on-topic]
- **Park et al.** — Diagnosing explainability of machine learning: transformer-based $PM_{2.5}$ estimation in Seoul. *Atmos Environ*, deposited 28 July. `10.1016/j.atmosenv.2026.122264`
- **Delgado Peralta et al.** — Improving $PM_{2.5}$ simulations in metropolitan São Paulo through PMF-based emission-inventory adjustment and local road transport emissions. *Atmos Environ*, deposited 27 July. `10.1016/j.atmosenv.2026.122249`
- **Faggi et al.** — Heatwaves and particulate matter increases: an Italian case study. *Atmos Environ*, deposited 27 July. `10.1016/j.atmosenv.2026.122246`
- **Yu et al.** — Indoor particulate matter and associated microplastic exposure in the educational environment of Shenzhen. *Atmos Pollut Res*, deposited 30 July. `10.1016/j.apr.2026.103162`
- **Saygin et al.** — Source identification and seasonal changes of trace metals, protein content and oxidative potential of fine and coarse particulates from Sanliurfa after the Kahramanmaras earthquakes. *Atmos Pollut Res*, deposited 31 July. `10.1016/j.apr.2026.103164`

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Cai et al.	$PM_{2.5}$ as the dominant air pollutant associated with suicide among people living with HIV/AIDS: province-wide case-crossover	*Atmos Environ*	Metadata only (no abstract)	$PM_{2.5}$ / PM_{10}	China
Daoud et al.	Air pollution and suicide mortality in the UK: implications for environmental and mental health policy	*Environ Sci Pollut Res Int*	Ecological panel, fixed effects	$PM_{2.5}$ (modelled, DEFRA)	UK
Zhang et al.	Association of $PM_{2.5}$ and its components with cognitive dysfunction among rural older adults	*Neurotoxicology*	Cross-sectional (cohort baseline)	$PM_{2.5}$, BC, OM, NH4, SO4, NO3 (TAP)	China
Reproductive & developmental					
Bello et al.	Prenatal exposure to $PM_{2.5}$ and infant health: evidence from Quebec	*Econ Hum Biol*	Registry cohort, FE models	$PM_{2.5}$ (monitor-assigned)	Canada
Pai et al.	Household incense burning and small for gestational age risk in newborn infants	*Pediatr Neonatol*	Prospective birth cohort	Incense PM (indoor, unspeciated)	Taiwan
Respiratory & allergic					
Lu et al.	Antagonistic effects of $PM_{2.5}$ and ozone on respiratory morbidity: multi-pollutant interaction, Pearl River Delta	*Atmos Environ*	Metadata only (no abstract)	$PM_{2.5}$ / PM_{10}	China
Mechanistic toxicology					

First author	Focus	Journal	Design	Metric	Region
Mishra et al.	Aerosol oxidative potential and reactive species predicted with a chemical kinetics model	*Atmos Chem Phys*	Kinetic model + field validation	PM composition, OP-DTT / OP-AA	France / UK
Saygin et al.	Source identification and seasonal change of trace metals, protein content and oxidative potential of fine and coarse particulates after the Kahramanmaras earthquakes	*Atmos Pollut Res*	Metadata only (no abstract)	$PM_{2.5}$ / PM_{10}	Turkiye
Occupational & indoor					
Yu et al.	Indoor particulate matter and associated microplastic exposure in the educational environment of Shenzhen	*Atmos Pollut Res*	Metadata only (no abstract)	$PM_{2.5}$ / PM_{10}	China
Zhou L et al.	Indoor burning of Arabian incense generates abundant ultrafine particles with strong oxidative potential	*Atmos Chem Phys*	Chamber emission characterisation	UFP number + PM mass, DTT/cellular OP	Chamber (Gulf practice)
van Rheenen et al.	Microbial contamination of air conditioning and humidifying systems in hospitals and the built environment	*J Hosp Infect*	Mapping review	Bioaerosol (bacteria, fungi, protozoa)	Multi-country
Sensing, forecasting & instrumentation					
Kumpika et al.	Correction of humidity and sensor aging effects in low-cost $PM_{2.5}$ light-scattering sensors	*Environ Pollut Bioavailab*	Field co-location + chamber	$PM_{2.5}$ (PMS7003 vs BAM)	Thailand
Le et al.	Operational performance of the Vaisala CL61 ceilometer for atmospheric profiling	*Atmos Meas Tech*	Multi-instrument field evaluation	Attenuated backscatter, depolarisation ratio	Finland / Germany
Mohapatra et al.	IoT-machine learning proof-of-concept framework for real-time urban particulate matter prediction	*IEEE Access*	Proof-of-concept deployment	PM1 / $PM_{2.5}$ / PM_{10} (PMS7003)	India
Nurseitov et al.	Design, deployment and field evaluation of a low-cost IoT monitoring network for urban particulate matter	*Information*	Network deployment + field evaluation	$PM_{2.5}$ / PM_{10} (ZPHS01B)	Kazakhstan
Park et al.	Diagnosing explainability of machine learning: transformer-based $PM_{2.5}$ estimation in Seoul	*Atmos Environ*	Metadata only (no abstract)	$PM_{2.5}$ / PM_{10}	Republic of Korea
Qian et al.	Enhancing accuracy of indoor air quality sensors via automated machine learning calibration	*Atmos Meas Tech*	Chamber co-location, AutoML calibration	$PM_{2.5}$ (two LCS models vs reference)	Chamber
Reis	Low-power embedded sensor node for environmental monitoring with on-board machine-learning inference	*Sensors*	Simulation (system-level)	PM + gas, multi-parameter node	Simulation
Silva et al.	Low-cost sensors for indoor air quality monitoring: systematic review of accuracy, applications and limitations	*J Air Waste Manag Assoc*	Systematic review (PRISMA)	PM (OPC) and CO2 (NDIR)	Multi-country
Zheng et al.	Augmented transformer framework for $PM_{2.5}$ chemical composition via one-step LDI single-particle aerosol mass spectrometry	*Atmos Environ*	Metadata only (no abstract)	$PM_{2.5}$ / PM_{10}	China
Exposure assessment & modelling					
Blanco et al.	Influence of on-road mobile monitoring design on ultrafine particle exposure models and cognitive health inferences	*J Expo Sci Environ Epidemiol*	Design resampling + cohort analysis	UFP particle number concentration	United States
Delgado Peralta et al.	Improving $PM_{2.5}$ simulations in metropolitan Sao Paulo via PMF-based emission inventory adjustment	*Atmos Environ*	Metadata only (no abstract)	$PM_{2.5}$ / PM_{10}	Brazil
Saeedi et al.	Feature selection and cross-validation strategy in land-use regression for mobile-monitored ultrafine particles	*Environ Sci Technol*	LUR model comparison	Ultrafine particle number (mobile)	Canada
Zhou M et al.	Surface $PM_{2.5}$ air pollution in 2022 India: emission updates, WRF-Chem evaluation, and source attribution	*Atmos Chem Phys*	CTM simulation + source attribution	$PM_{2.5}$ (WRF-Chem, speciated)	India
Burden, policy & mitigation					
Faggi et al.	Heatwaves and particulate matter increases: an Italian case study	*Atmos Environ*	Metadata only (no abstract)	$PM_{2.5}$ / PM_{10}	Italy

Provenance and method

Windows. The PubMed and Europe PMC legs cover **1 August 2026** alone, contiguous with the 31 July issue — no gap. PubMed was queried on `[Date - Entry]` along two independent axes (health, instrumentation): 11 records in the window, 5 retained. Europe PMC `CREATION_DATE:[2026-08-01 TO 2026-08-01]` returned 2, both already published in the 31 July issue and dropped as duplicates. arXiv returned one record inside the recency screen (Dirichlet-mixture censoring methods), rejected for having no PM application. OpenAlex date filtering remains behind a paid plan and was not used.

New this run: the Crossref by-journal leg. The 31 July issue diagnosed the sensing blackout and prescribed a Crossref feed keyed on `created` for the journals PubMed does not index. That leg is now implemented in `harvest.py` (`crossref_journal`, `crossref_sensing`) over eight ISSNs: *Atmos. Meas. Tech.*, *Atmos. Chem. Phys.*, *Aerosol Sci. Technol.*, *Atmos. Environ.*, *J. Aerosol Sci.*, *Atmos. Pollut. Res.*, *Aerosol Air Qual. Res.*, *Environ. Sci.: Atmos.* A *global* `from-created-date` filter is useless for entry-date windowing — continuous re-indexing returns works back to 2007 — but scoped to a single journal ISSN it returns that journal's genuinely new deposits, verified today. Its first run back-swept **26 July–1 August** and returned **55 deposits, 12 PM-relevant**, all 12 carried here (4 with abstracts, 8 metadata-only). They are dated to this issue because 1 August is when the leg first surfaced them — the same convention applied to the Europe PMC records in the 31 July issue. **The issue's window is therefore not uniform across sources**, which is stated rather than hidden.

Consensus leg. Eight further sensing and exposure-modelling records came from Consensus sweeps on calibration and drift, co-location against reference monitors, network siting, edge inference and mobile monitoring. Consensus has no entry-date filter, so these are screened against `state/seen.json` rather than by date; two hits (Ding, Nguyen) were dropped as already published on 29 July. DOIs were recovered by Crossref bibliographic matching, since Consensus returns none.

Screening. 25 records retained, **15 rejected with a logged reason** in `state/rejected.jsonl`: 4 duplicates across sources, 3 gas-phase-only instrumentation, 2 non-atmospheric particulates (pharmaceutical surface cleanliness, marine sediment resuspension), 1 catalysis review, 1 heat-mortality study in which PM enters only as a covariate, 1 therapeutic-nebuliser deposition study, 1 generic statistical method with no PM application, 1 with no retrievable abstract from any connector, 1 formaldehyde hematotoxicity review.

Trial watch. The ClinicalTrials.gov sweep on filtration interventions added three sham-controlled trials to `state/trials.json` (NCT05867381, NCT05718245, NCT05016271); they are surfaced in the weekly rollup, not here.

Subtopic, design, particle-metric and geography labels were assigned from title, abstract, keywords and MeSH terms; assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary. Effect estimates in the forest plot are transcribed verbatim from abstracts. Every DOI was verified and resolved before publication. Content derived from PubMed, Europe PMC and Crossref metadata; abstracts remain the copyright of their respective publishers.